

Embedded system (IoT) Developer Task

Task Description:

Your objective is to create display driver software. The display has 7-segment values and icons (like Wi-Fi, Bluetooth, Ice, Fire, Humidity, and Sun).

The driver software should have the following functionalities:

1. A function to initialize the display.

```
displayInit();
```

2. A function to clear the whole screen.

```
displayClear();
```

3. A function to enable or disable a specified Icon.

- i. `icon(BLE_ICON, true);` // Enable: show icon on the display.
- ii. `icon(BLE_ICON, false);` // Disable: Bluetooth icon disappears from the display.

4. A function to print values on the display.

- i. `printTemperature(130.5, degF);` // print 130.5 °F on the display and do this for the degC too.
- ii. `printHumidity(35.3);` // print 35.3% on the display.

5. A function to Set Battery Level.

```
printBattLevel(60);
```

 // print 60% of Battery Level on the display.

Hint: A single bar in the battery icon represents 20% of Battery Level.

Note: We provided the display driver datasheet and Screen pinout. (All the information you need to design the driver software).

Driver Features:

1. **Display Initialization:** This Function will set up the display.

Like Display driver mode, internal voltage adjustment, blink frequency, etc.

2. **Enable / Disable Icons:** A function that allows you to display specified icons on the display and remove the icons if the user wants.
3. **Print numbers:** 3 functions for printing values of Temperature, Humidity, and Battery Level on display.
4. **Clear Display:** A function that clears the whole screen.

Requirements:

- You can use any microcontroller of your choice to design the driver software. (Good if you use ESP32 and ESP-IDF).
- The API function should have minimal input arguments (if possible).
- Make sure that the driver's software Compiles without errors and works on the Hardware. All the features are mentioned above.

Submission:

Please provide the .cpp and .h files of your display driver software, and the example application that uses this driver.

Evaluation:

Your submission will be evaluated based on the following criteria:

1. **Functionality:** Does the driver's software meet the specified requirements and features?
2. **User Interface:** Are the driver software functions easily usable by other developers in their application?
3. **Code Quality:** Is the code well-structured, documented, and maintainable?
4. **Testing:** Have you implemented basic testing to ensure core functionality?

Note: This task is designed to evaluate your technical skills in Device driver design and Implementation. It shows your ability to read the data sheet and extract useful information from it.

Good luck with the task. We shall look forward to reviewing your submission!